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Theoretical Physics

Quantum Field Theory II

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Introduction

In these notes, we will explore the fascinating world of Quantum Field Theory (QFT). QFT is a fundamental framework in physics that combines classical field theory, special relativity, and quantum mechanics. It provides a powerful language for describing the behavior of particles and fields at the most fundamental level. We assume are going to assume that the reader has a basic understanding of quantum mechanics and special relativity, as well as some familiarity with classical field theory and quantum field theory of free fields (Which will anyway be reviewed in the first chapter).

There exist different motivations for studying QFT: it provides a framework for describing the interactions between particles and fields, and it has been successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of condensed matter systems. Additionally:

- QFT is the language of the Standard Model of particle physics, which describes the fundamental particles and their interactions (except for gravity); it is used for the quantization of systems with an **infinite number of degrees of freedom**, such as fields; it is essential for understanding the behavior of particles at high energies and small distances, where quantum effects become significant; and it provides a framework for describing the behavior of particles in curved spacetime, which is important for understanding phenomena such as black holes and the early universe.
- It is a successful framework for describing the behavior of particles and fields at the most fundamental level, incorporating both quantum mechanics and special relativity:
 - At high energies we can witness **particles creation and annihilation**, which cannot be described by non-relativistic quantum mechanics; QFT provides a framework for describing these processes, in an infinite-dimensional system, where the number of particles is not fixed.
 - QFT is a relativistic theory, which means that it is consistent with the principles of special relativity, implmenting **causality** and **Lorentz invariance**; this is essential for describing the behavior of particles at high energies and small distances, where relativistic effects become significant. So two events that are spacelike separated cannot influence each other

$$(x - y)^2 = (x^0 - y^0)^2 - \sum_{i=1}^3 (x^i - y^i)^2 < 0 \implies [\mathcal{O}(x), \mathcal{O}(y)] = 0,$$

thus the events are causally disconnected, ensuring causality.

- Particles are described as excitations of underlying fields, which can be created and annihilated; this allows for a more complete description of particle interactions and the behavior of particles at high energies. Examples of particles and underlying fields include:
 - **Scalar field:** $\phi(x) \implies$ spin-0 particles (e.g., Higgs boson).
 - **Spinor field:** $\psi(x) \implies$ spin-1/2 particles (e.g., electrons, quarks).
 - **Vector field:** $A_\mu(x) \implies$ spin-1 particles (e.g., photons, W and Z bosons).
- The **Standard Model** of particle physics is a quantum field theory that describes the fundamental particles and their interactions (except for gravity); it has been successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of condensed matter systems. It is a quantum field theory that describes the electromagnetic, weak, and strong interactions, through the exchange of gauge bosons among fields of spin 0, 1/2, and 1 particles.
- The inclusion of gravity in a quantum field theory framework is an open problem in theoretical physics, but can be approached by inserting the metric tensor of general relativity

$$g^{\mu\nu} = \eta^{\mu\nu} + h^{\mu\nu}.$$

While there are approaches to quantizing gravity, such as string theory and loop quantum gravity, a complete and consistent theory of quantum gravity has not yet been developed. Quanta of the gravitational field are called gravitons, which are hypothetical spin-2 particles that mediate the gravitational interaction, but they have not been observed experimentally. These notes will not cover quantum gravity, but it is an important area of research in theoretical physics which sometimes will be mentioned throughout the notes.

The main goal of these notes is to provide a comprehensive understanding of interacting quantum field theories, which are essential for describing the behavior of particles and fields at the most fundamental level, as well as mathematical methods for making theoretical predictions on interactions among particles. We will cover a wide range of topics, including:

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- The quantization of free fields, which provides the foundation for understanding the behavior of particles and fields in quantum field theory.
- The construction of interacting quantum field theories, which are essential for describing the behavior of particles and fields at the most fundamental level.
- The use of perturbation theory and Feynman diagrams to calculate scattering amplitudes and other physical observables in quantum field theory.
- The renormalization of quantum field theories, which is necessary to make sense of the infinities that arise in perturbation theory and to extract meaningful physical predictions from the theory.
- The application of quantum field theory to the Standard Model of particle physics, which describes the fundamental particles and their interactions (except for gravity).

QFT is one of the most successful and powerful frameworks in physics, and it has been instrumental in our understanding of the fundamental nature of the universe. This theory has been tested and

confirmed by a wide range of experiments, some of which tested the predictions of QFT to an extraordinary degree of precision. For example, the anomalous magnetic moment of the electron has been measured to an accuracy of one part in a trillion ($\sim 10^{-13}$), and the predictions of QFT have been confirmed to this level of precision. Additionally, the discovery of the Higgs boson at the Large Hadron Collider in 2012 provided further confirmation of the predictions of QFT and the Standard Model.

Another essential aspect of these notes is to provide a comprehensive understanding of the importance of symmetries in quantum field theory, which play a crucial role in determining the behavior of particles and fields. We will explore the concept of gauge symmetries, which are local symmetries that arise in quantum field theories and are responsible for the interactions between particles and fields. We will also discuss the role of global symmetries, which are symmetries that do not depend on spacetime coordinates, and their implications for the behavior of particles and fields. In the end we will argue that *a QFT is essentially defined by its fields and its symmetries* (statement that will be made more precise in the end of the notes), and we will see how symmetries can be used to constrain the behavior of particles and fields, and to make predictions about their interactions.

For a more comprehensive introduction to quantum field theory, we recommend the following textbooks:

- **An Introduction to Quantum Field Theory** by Michael E. Peskin and Daniel V. Schroeder: This is a widely used textbook that provides a comprehensive introduction to quantum field theory, covering both the theoretical foundations and practical applications of the subject. TODO: check and change in the end
- **Quantum Field Theory** by Mark Srednicki: This textbook offers a clear and concise introduction to quantum field theory, with an emphasis on the physical intuition behind the mathematical formalism.
- **Quantum Field Theory in a Nutshell** by A. Zee: This book provides a more informal and intuitive introduction to quantum field theory, making it accessible to a wider audience while still covering essential topics.
- **The Quantum Theory of Fields** by Steven Weinberg: This three-volume series is a comprehensive and rigorous treatment of quantum field theory, written by one of the pioneers in the field. It is suitable for advanced students and researchers who want a deep understanding of the subject.

For more information about the notation and conventions used in these notes, please refer to the appendix, where we provide a detailed explanation of the symbols and mathematical conventions that we will use throughout the notes (Appendix A).

Prerequisites: *Familiarity with the Lagrangian and Hamiltonian formalisms of Classical Mechanics, standard Quantum Mechanics, Special Relativity and the basics of Quantum Field Theory (free fields) is required.*

1 | Free QFT Review

A free quantum field theory does not account for interactions between particles. It describes particles that do not interact with each other, and the fields that represent them evolve independently. Free QFTs are often used as a starting point for understanding more complex interacting theories, as they provide a simpler framework to study the properties of quantum fields and particles.

The lagrangian for a free quantum field theory typically consists of kinetic terms and mass terms for the fields, without any interaction terms: it can only contain **quadratic terms** in the fields, without any higher-order interactions. Quanta of the free fields correspond to particles that do not interact with each other, and their dynamics are governed by the **free propagators** of the theory, which describe how particles propagate through spacetime without any interactions: propagators are the Green's functions of the free theory, and they determine how particles propagate from one point to another in spacetime with determined probability amplitudes.

We will review the free quantum field theories for different types of particles, including scalar fields (spin 0), fermionic fields (spin 1/2), and gauge fields (spin 1). We will also discuss the symmetries of these theories and how they lead to conservation laws through Noether's theorem. This review will provide a foundation for understanding more complex interacting quantum field theories, and it will also highlight the differences between free and interacting theories, as well as the role of symmetries in quantum field theory.

1.1 | Spin 0

For a real scalar field $\phi(x)$, the free Lagrangian is given by

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2 \phi^2, \quad (1.1.1)$$

where m is the mass of the scalar particle. The equation of motion derived from this Lagrangian is the **Klein-Gordon equation**:

$$(\square + m^2)\phi = 0, \quad (1.1.2)$$

where $\square = \partial_\mu \partial^\mu$ is the d'Alembertian operator. The solutions to this equation describe free scalar particles with mass m . This is a classical equation of motion, having the same role of Maxwell's equations in classical electromagnetism, but for a scalar field $\phi(x)$ instead of a vector field $A^\mu(x)$.

The idea of quantum field theory is to promote the classical field $\phi(x)$ to a quantum operator $\hat{\phi}(x)$ that acts on a Hilbert space of states. The quantization procedure involves imposing commutation relations on the field operators, which leads to the creation and annihilation of particles. The free scalar field can be expanded in terms of creation and annihilation operators, which allows us to describe the particle content of the theory. Indeed, we can see how decomposing the field into its Fourier modes leads to the interpretation of the quanta of the field as particles with specific momenta and energies:

$$\phi(x) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} e^{i\mathbf{k}\cdot\mathbf{x}} \tilde{\phi}(t, \mathbf{k}),$$

where each mode can be computed by using this ansatz in the Klein-Gordon equation in momentum space, leading to the dispersion relation of an harmonic oscillator represented by on-shell particles with energy $E_{\mathbf{k}} = \omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}$:

$$(\partial_t^2 + \omega_{\mathbf{k}}^2) \tilde{\phi}(t, \mathbf{k}) = 0,$$

showing how the free scalar field can be interpreted as a collection of independent harmonic oscillators, each corresponding to a different momentum mode.

In order to quantize the free scalar field, we have to promote the fields themselves to operators and impose the canonical commutation relations, which leads to the creation and annihilation of particles. Before showing how to do that, it is crucial to note that $\hat{\phi}(x)$ depends on both space and time, enforcing the Heisenberg picture of quantum mechanics, where operators evolve in time while states remain fixed. The canonical commutation relations for the scalar field are given by

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y}),$$

where $\hat{\pi}(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi(x))} = \partial_0 \hat{\phi}(x)$ is the conjugate momentum operator. This constraint along with reality of the field $\hat{\phi}(x) = \hat{\phi}^\dagger(x)$ allows us to express the field operator in terms of **creation and annihilation operators**:

$$\hat{\phi}(x) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-ik \cdot x} + \hat{a}_{\mathbf{k}}^\dagger e^{ik \cdot x} \right),$$

where $k \cdot x = \omega_{\mathbf{k}} t - \mathbf{k} \cdot \mathbf{x} = k^\mu x_\mu$, and $\omega_{\mathbf{k}} = k^0$. The creation and annihilation (ladder) operators satisfy the following commutation relations

$$\begin{cases} [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta^3(\mathbf{k} - \mathbf{k}'), \\ [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}] = [\hat{a}_{\mathbf{k}}^\dagger, \hat{a}_{\mathbf{k}'}^\dagger] = 0. \end{cases}$$

Note that while the field operator $\hat{\phi}(x)$ is Hermitian and time dependent, the creation and annihilation operators are not, and they are responsible for creating and destroying particles in the quantum field theory. The **vacuum state** $|0\rangle$ is defined as the state that is annihilated by all annihilation operators:

$$\hat{a}_{\mathbf{k}} |0\rangle = 0 \quad \forall \mathbf{k},$$

while we can create one-particle states by acting with the creation operators on the vacuum, which adds one quanta of the field with momentum $\mathbf{k}$:

$$|\mathbf{k}\rangle = \sqrt{2\omega_{\mathbf{k}}} \hat{a}_{\mathbf{k}}^{\dagger} |0\rangle,$$

and more generally, we can create multi-particle states by acting with multiple creation operators on the vacuum, which leads to the **Fock space** of the theory. With this definition of single particle states, we can compute the inner product between two single particle states, which is given by

$$\langle \mathbf{k} | \mathbf{k}' \rangle = 2\omega_{\mathbf{k}} \delta^3(\mathbf{k} - \mathbf{k}'),$$

showing that the single particle states are orthogonal and normalized with respect to the relativistic normalization convention. The normalization $\sqrt{2\omega_{\mathbf{k}}}$ is arbitrary, but important for ensuring that the probabilities computed in the theory are consistent with relativistic invariance.

Remark. *To find a Lorentz invariant measure, we can start from the Minkowski space measure d^4k and impose the on-shell condition $k^2 = m^2$ using a delta function along with a step function to ensure that we only consider positive energy solutions:*

$$\int d^4k \delta(k^2 - m^2) \theta(k^0) = \int \frac{d^3\mathbf{k}}{2\omega_{\mathbf{k}}}.$$

We have then found the Lorentz invariant measure $\frac{d^3\mathbf{k}}{2\omega_{\mathbf{k}}}$ for integrating over the momentum space of on-shell particles, which is crucial for ensuring that the theory is consistent with special relativity. The previous normalization of the single particle states is consistent with this Lorentz invariant measure, since it ensures that the inner product between single particle states is Lorentz invariant, and it also allows us to compute probabilities and cross sections in a way that is consistent with relativistic invariance:

$$\int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{k}}} |\mathbf{k}\rangle \langle \mathbf{k}| = 1,$$

which is the completeness relation for the single particle states in the theory. This ensures that we can express any state in the Fock space as a superposition of single particle states, and it also allows us to compute transition amplitudes and probabilities in a way that is consistent with relativistic invariance.

1.1.1 | Feynman's Propagator

1.1.2 | Noether's Theorem

1.2 | Spin 1/2

1.3 | Spin 1

2 | Interacting Fields

Free theories are a good starting point to understand the basic concepts and techniques of quantum field theory, but to capture the full complexity of the physical world, we need to introduce interactions between fields. Interactions allow us to describe processes such as scattering, particle decays, and the formation of bound states, which are essential for understanding the behavior of matter and energy at a fundamental level. In a free theory, the fields do not interact with each other, and the equations of motion are linear: they can exhibit *at most quadratic terms in the fields* (in the lagrangian $\mathcal{L}$). We can then build interacting theories by adding non-linear terms to the Lagrangian, which will lead to non-linear equations of motion. These non-linearities are what give rise to the rich phenomenology of interacting quantum field theories.

The most important tools in the developing and understanding of interacting quantum field theories are:

- **Dimensional analysis**, which allows us to determine the relevance of different interaction terms at different energy scales, and to classify interactions as relevant, irrelevant, or marginal.
- **Renormalization**, which is a systematic procedure to deal with the infinities that arise in perturbative calculations of interacting theories, and to extract meaningful physical predictions from them.
- **Symmetry principles**, which play a crucial role in constraining the form of the interactions and in determining the properties of the particles and fields in the theory. For example, gauge symmetries lead to the introduction of gauge fields, which mediate the fundamental forces of nature.

Moreover **locality** is a fundamental principle in quantum field theory, which states that interactions should occur at the same point in spacetime. This principle ensures that the theory respects causality and allows us to construct a consistent framework for describing the dynamics of fields and particles:

$$\mathcal{L}(x) = \mathcal{L}(\phi(x), \partial_\mu \phi(x)), \quad \mathcal{L}(x) \neq \mathcal{L}(\phi(x+y), \partial_\mu \phi(x+y)),$$

and the interaction terms we will introduce will be local, meaning that they will involve products of fields and their derivatives evaluated at the same spacetime point x ; this ensures that the theory is consistent with the principles of special relativity and quantum mechanics, and it will guide us in constructing the interaction terms in the Lagrangian.

In the next section we will start by performing a dimensional analysis of the fields and the interaction terms, which will help us to understand the behavior of the theory at different energy scales

and to identify the most relevant interactions for describing physical phenomena. This will set the stage for our subsequent discussions on renormalization and symmetry principles in interacting quantum field theories.

2.1 | Dimensional Analysis

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2.1.1 | Examples of Interaction Terms

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ϕ^3 Interaction

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ϕ^4 Interaction

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ϕ^n General Interaction

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Complex Scalar Field Interactions

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Quantum Electrodynamics (QED) Interaction

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Yukawa Theory Interaction

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Scalar QED (sQED) Interaction

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2.1.2 | Master Plan

In the next sections, we will introduce interactions in the theory by adding interaction terms to the Lagrangian. The simplest way to do this is to add a term that is a polynomial in the field ϕ , such as $\lambda\phi^4$ or $g\phi^3$. These interaction terms will lead to non-linear equations of motion, which will make the theory more complicated to solve. However, they will also allow us to describe a wider range of physical phenomena, such as scattering processes and particle decays.

More precisely, we are going to:

- Define useful concepts and observables in the theory, such as **cross sections** and **decay rates**.
- Express these observables in terms of the **S-matrix**, which encodes the transition amplitudes between initial and final states in scattering processes: the matrix elements of a generic operator represent the probability amplitude of measuring a specific eigenvalue, corresponding to a transition from an initial state $|i\rangle$ to a final state $|f\rangle$.
- We will use theorems and techniques developed in a QFT framework, such as the **LSZ reduction formula**, to relate the S-matrix elements to correlation functions of the fields or their Fourier transforms, which are the fundamental objects of study in QFT. For example, correlation functions of the form

$$\langle 0 | T \{ \phi(x_1) \phi(x_2) \cdots \phi(x_n) \} | 0 \rangle ,$$

will show how the fields interact at different points in spacetime, and can be used to compute scattering amplitudes and other physical observables.

- We will develop in the end procedures and techniques to compute these correlation functions from an interacting lagrangian in **perturbation theory**; using **Feynman diagrams** and **Feynman rules** we will be able to systematically calculate the contributions of different interaction terms to the physical observables.

2.2 | Scattering Processes

Scattering processes are currently the most important tool to probe the fundamental interactions of nature, and thus to test our theories of particle physics. They let us infer and understand the properties of particles and their interactions at a microscopic level by analyzing the outcomes of experiments and collisions at macroscopic distances.

We are forced to cite some of the most important results of scattering theory, as well as experiments, which is a vast and complex subject, in order to understand its importance in particle physics. Some of the most important results and experiments in scattering theory include:

- **Rutherford scattering**, which was the first experiment to probe the structure of the atom and led to the discovery of the nucleus. Scattering alpha particles off a thin gold foil, Rutherford observed that most of the particles passed through the foil with little deflection, while a small fraction were scattered at large angles, leading to the conclusion that the atom has a small, dense nucleus at its center. This experiment led to our modern understanding of the atomic structure of matter at a subatomic level, and it was a crucial step in the development of quantum mechanics and quantum field theory.
- **Lawrence Berkeley National Laboratory**, which was the first particle accelerator and allowed for the discovery of many new particles and interactions: in 1931 Lawrence was the first to accelerate particles to relativistic energies, firstly only around 80 keV, and then with Livingston he built the first cyclotron, which was able to accelerate protons to energies of about 1.22 MeV, leading to the discovery of the neutron in 1932 by James Chadwick.
- **Large Hadron Collider (LHC)**, which is the largest and most powerful particle accelerator in the world, located at CERN, and has been instrumental in the discovery of the Higgs boson in 2012, as well as in probing the properties of the Standard Model and searching for new physics beyond it. It has a circumference of about 27 km and can accelerate protons to energies of up to 7 TeV per beam, allowing for collisions at unprecedented energies and luminosities (actually around 13 TeV, with future upgrades planning further increases).
- **Future colliders**, such as the proposed Future Circular Collider (FCC) at CERN, which aims to reach even higher energies and luminosities than the LHC, and could potentially discover new particles and interactions that are beyond the reach of current experiments. Basically all we know about particle physics and the fundamental interactions of nature has been inferred from the observations and measurements of scattering processes, either in natural phenomena (such as cosmic rays) or in controlled experiments.

Such great energies and luminosities allow us to probe the fundamental interactions of nature at a microscopic level, and to test our theories of particle physics with high precision: the **indetermination principle** allows us to probe smaller and smaller distances by increasing the energy of the particles, thus allowing us to explore the structure of matter and the fundamental forces at a deeper level. By analyzing the outcomes of scattering experiments, we can infer the properties of particles and their interactions, and test our theoretical models against experimental data.

Also the **decay rate** of unstable particles is an important observable that can be measured in experiments and computed in theory, providing insights into the properties of the particles and their interactions. In these decay processes, an unstable particle decays into other smaller and lighter particles, and the decay rate quantifies the probability per unit time that this decay will occur.

By studying the decay rates of particles, we can gain insights into their properties, such as their masses, lifetimes, and interaction strengths, and test our theoretical models against experimental data.

2.2.1 | S-Matrix

We prepare a state $|i, t_i\rangle$ at an initial time t_i which will transition after some interaction (or decay, scatter...) into a state $|f, t_f\rangle$, measured at a final time t_f with a probability given by the square of the amplitude $|\langle f, t_f | i, t_i \rangle|^2$. Here we can make the following observations:

- The initial and final states are defined in the limit of infinite past and future, where the particles are free and non-interacting. This is a common assumption in scattering theory, as it allows us to define **asymptotic states** that can be used to compute scattering amplitudes and cross sections.
- The process happens on microscopic time and length scales, while being measured on macroscopic scales, thus we can treat the interaction as an instantaneous event, letting us ignore the detailed time dependencies of the process and focus on the transition between the initial and final states.
- The transition from the initial to the final state can be described by a unitary operator called the **S-matrix**, which encodes the **transition amplitudes** between the asymptotic states.

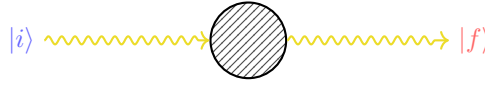


Figure 2.1: Schematic representation of a generic process, where an initial state $|i\rangle$ evolves into a final state $|f\rangle$ through some interaction described by the S-matrix. We are considering asymptotic states, which are defined in the limit of infinite past and future, where the particles are free and non-interacting. The S-matrix encodes the transition amplitudes between these asymptotic states describing the scattering process happening on microscopic time scales.

We typically use **Heisenberg picture**, so if we want to compute the amplitude of a determined process, we can write it as

$$\langle f | \hat{S} | i \rangle = \langle f, \infty | i, -\infty \rangle,$$

where $|i\rangle$ and $|f\rangle$ are the asymptotic states of the system (time independent and directly related to the free particle states), respectively. *The S-matrix element $\langle f | \hat{S} | i \rangle$ gives us the amplitude for the transition from the initial state $|i\rangle$ to the final state $|f\rangle$ due to the interaction described by the S-matrix itself.* The S-matrix is a powerful tool in quantum field theory, as it allows us to compute **scattering amplitudes** and **cross sections** for a wide range of processes, and to make predictions that can be tested against experimental data. By analyzing the S-matrix elements for different processes, we can gain insights into the properties of particles and their interactions, testing our theoretical models against experimental observations.

2.3 | Cross Sections

If we analyze a scattering process, such as the Rutherford scattering, we can see how a beam of particles is scattered by a target; in the case of Rutherford scattering, it were alpha particles being scattered by a thin gold foil, where the alpha particles are much smaller than the nucleus of the gold atoms, thus we can treat the nucleus as a still sphere and the alpha particles as point-like particles. We will make the same assumption for the following scattering processes, thus treating the particles as point-like and the interactions as instantaneous events.

In this case, we can define the **cross section** of the target, which quantifies the probability of a scattering event occurring when a particle from the beam interacts with the target. The cross section is an effective area that represents the likelihood of a scattering event, and it can be computed using the S-matrix elements for the process.

The **geometric cross section** of the target, which encodes the effective area of the target for scattering, is given by

$$\sigma \equiv \pi R^2, \quad (2.3.1)$$

where R is the radius of the target. One can compute the number of particles that are scattered by the target, which is given by

$$N = n_B \cdot \sigma, \quad n_B = \frac{N_B}{A},$$

where n_B is the number density of the beam, N_B is the number of particles in the beam, and A is the area of the target. The cross section σ can be interpreted as an effective area that quantifies the likelihood of a scattering event occurring when a particle from the beam interacts with the target. The power of this approach resides in its simplicity: with few assumptions it allows us to compute the probability of scattering events for a wide range of processes, while linking macroscopical observables, such as the number of scattered particles, to the microscopic dynamics of the interactions encoded in the S-matrix elements.

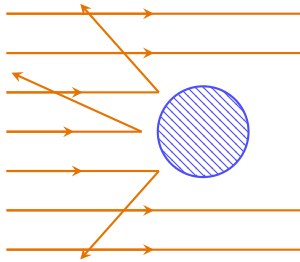


Figure 2.2: Schematic representation of a scattering process, where a beam of particles (orange arrows) is scattered by a target (blue circle). The geometric cross section of the target is given by $\sigma = \pi R^2$, with R being the radius of the target, while the number of scattered particles depends on the interaction probability. We are going to generalize this model to quantum field theory, computing the probability of scattering processes and expressing them in terms of the S-matrix elements.

Now we want to take this model and generalize it to quantum field theory, where we can compute the probability of scattering processes; we want to extend the model

- for any type of scattering process, or interaction;
- for a probability of scattering, not just a number of particles.

2.3.1 | Impact Parameter

If we assume a beam of particles along the z -axis, we can define the **impact parameter** $\mathbf{b}$, which represents the perpendicular distance among the trajectory of the incoming particle and the center of the target:

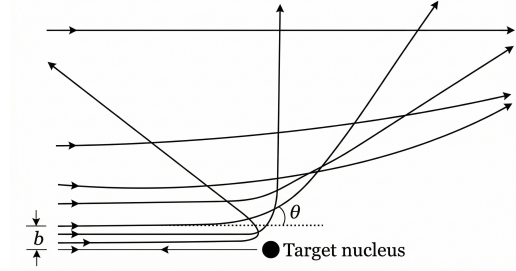
$$N = \int d^2\mathbf{b} n_B P(\mathbf{b}) = n_B \int d^2\mathbf{b} P(\mathbf{b}),$$

thus we can now define the **cross section** as

$$\sigma = \int d^2\mathbf{b} P(\mathbf{b}), \quad (2.3.2)$$

where $P(\mathbf{b})$ is the probability of scattering for a given impact parameter $\mathbf{b}$, generalizing the concept of a geometric cross section. This expression allows us to compute the cross section for a scattering process by integrating the probability of scattering over all possible impact parameters, thus providing a more general and flexible framework for analyzing scattering processes in quantum field theory.

Figure 2.3: Schematic representation of a scattering process in terms of the impact parameter $\mathbf{b}$: the probability of scattering can be expressed as a function of the impact parameter, allowing us to compute the cross section by integrating the probability of scattering over all possible impact parameters.



We can also express it in its **differential form**, which ideally counts only a subset of events, with respect to the **solid angle** Ω as

$$\frac{d\sigma}{d\Omega} = \int d^2\mathbf{b} \frac{P(\mathbf{b})}{d\Omega}, \quad (2.3.3)$$

where we are considering only the events that are scattered in a specific direction defined by the solid angle $d\Omega$, thus providing a more detailed description of the scattering process. Alternatively, we can express it with respect to the measure of the space of **final states** $|f\rangle$ as

$$\frac{d\sigma}{df} = \int d^2\mathbf{b} \frac{P(\mathbf{b})}{df}. \quad (2.3.4)$$

The consequent step towards the computation of the cross section in QFT are:

- Express the probability of scattering $P(\mathbf{b})$ in terms of the S-matrix elements for the process

$$\langle f | \hat{S} | i \rangle = \langle f, \infty | i, -\infty \rangle,$$

which encode the transition amplitudes between the initial and final states.

- Integrate the probability of scattering over all possible impact parameters $\mathbf{b}$ to compute the cross section for the scattering process, using the expressions in equations (2.3.2), (2.3.3), or (2.3.4); then express the result in terms of observables that can be measured in experiments, such as the scattering angle or the energy or momenta of the scattered particles.

2.3.2 | Wave Packets and Regularization

Now one of the natural questions that arises is: what are the initial and final states $|i\rangle$ and $|f\rangle$ that we should use to compute the S-matrix elements for the scattering process? In typical scattering processes, the quantum dispersion of the particle momenta Δp is smaller than the experimental resolution, thus it makes sense to use **eigenstates of the momentum operator**, such as plane wave states. However, there are some issues:

- If the quantum dispersion of the particle momenta Δp is comparable to zero, for the Heisenberg uncertainty principle, the particle is completely delocalized in space

$$\Delta p \Delta x \geq \frac{\hbar}{2}, \quad \Delta p = 0 \implies \Delta x \rightarrow \infty,$$

thus it can interact with the target at any point in space, leading to divergences in the scattering amplitudes.

- We will show that S-matrix elements for plane wave states are proportional to delta functions, which leads to divergences when we compute the probability of scattering by taking the square of the amplitude: the probability of scattering is given by the square of the amplitude $\langle f | \hat{S} | i \rangle \sim \delta^{(4)}(p_i - p_f)$, but then

$$P \sim |\langle f | \hat{S} | i \rangle|^2 \sim |\delta^{(4)}(p_i - p_f)|^2 = \delta^{(4)}(p_i - p_f) \delta^{(4)}(0).$$

The term $\delta^{(4)}(0)$ is divergent, and it arises from the fact that we are using plane wave states, which are not normalizable and lead to divergences in the scattering amplitudes. This is a common issue in quantum field theory, and it requires regularization techniques to handle these divergences and obtain finite results for physical observables.

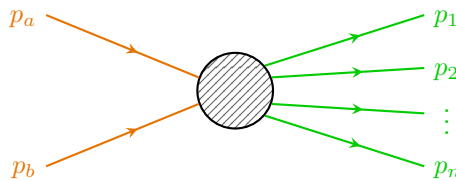
TODO: Add reference to book by schwarz.

One way to overcome the last problem is to consider a finite volume V and a finite time T , so that the delta function can be regularized and then take only at the end the limit $V \rightarrow \infty$ and $T \rightarrow \infty$. This is a common technique in quantum field theory to handle divergences and make sense of scattering amplitudes. By working in a finite volume and time, we can ensure that the calculations are well-defined and then take the appropriate limits to recover physical results. However, this approach can be obscure and cumbersome, and it can lead to complications in the calculations.

TODO: add reference to Peskin and Schroeder.

We will instead follow the Peskin and Schroeder approach, which is to consider the states as **wave packets** in the momentum space, which are **localized in both position and momentum space**, thus avoiding the issues with the delta function. This approach allows us to compute scattering amplitudes and cross sections without encountering the divergences associated with plane wave states. By using wave packets, we can ensure that the initial and final states are well-defined and that the scattering probabilities are finite and physically meaningful.

We can consider a general process in which the initial state is a two particle state, with momenta p_a and p_b , while the final state is a N particle state:



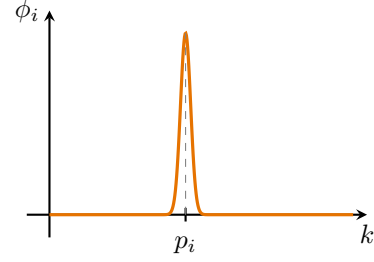
For incoming states we replace $|p_i\rangle$ with $|\phi_i\rangle$, which is a **wave packet** centered around the momentum p_i :

$$|p_i\rangle \rightarrow |\phi_i\rangle = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \phi_i(\mathbf{k}) |\mathbf{k}\rangle,$$

with

$$\langle\phi_i|\phi_i\rangle = \int \frac{d^3\mathbf{k}}{(2\pi)^3} |\phi_i(\mathbf{k})|^2 = 1,$$

remembering that $\langle\mathbf{k}|\mathbf{k}'\rangle = 2E_{\mathbf{k}} (2\pi)^3 \delta^{(3)}(\mathbf{k}-\mathbf{k}')$; all of this indicates that $\phi_i(\mathbf{k})$ is a very narrow normalized function centered around p_i .



Example (LHC collider). We can assume, very conservatively, that the LHC collider works with $\Delta p \sim 1$ MeV, which is unrealistic (since it would correspond to a resolution of about 10^{-6} , not realistic) and leads to a delocalization of the particles of about

$$\Delta x \sim \frac{\hbar}{2\Delta p} \sim \frac{1}{2 \text{ MeV}} \sim 100 \text{ fm} = 10^{-13} \text{ m},$$

which, compared to the size of the particles bunches in the LHC, which is of the order of $1 \mu\text{m} = 10^{-6} \text{ m}$, is negligible, thus we can safely treat the particles as localized in space, and thus we can use wave packets to compute the scattering amplitudes and cross sections without encountering divergences.

This is a common assumption in high-energy physics, with typically well-localized particles and small quantum dispersion of their momenta compared to the experimental resolution; this allows us to use wave packets to describe the initial and final states in scattering processes.

Now we are left with the task of computing the S-matrix elements for the scattering process using wave packets, and then integrating the probability of scattering over all possible impact parameters to compute the cross section for the scattering process. But how are we going to model the wave packets? We can consider different types of wave packets, such as **Gaussian wave packets**, which are commonly used in quantum mechanics and quantum field theory due to their nice mathematical properties and their ability to represent *localized states in both position and momentum space*. By choosing appropriate wave packet profiles, we can ensure that the initial and final states are well-defined and that the scattering probabilities are finite and physically meaningful.

Example (Gaussian wave packet). We can consider a Gaussian wave packet centered around p_i with a width σ :

$$\phi(\mathbf{k}) = N \exp \left[-\frac{1}{4} \left(\frac{(\mathbf{k} - \mathbf{p})^2}{\Delta p^2} \right)^2 - i\mathbf{k} \cdot \langle \mathbf{x} \rangle \right],$$

where N is a normalization constant, Δp is the width of the wave packet in momentum space, and $\langle \mathbf{x} \rangle$ is the average position of the wave packet in real space: using $\hat{x}^\mu = i \frac{\partial}{\partial p_\mu}$, one can check that the uncertainty principle is satisfied with $\Delta x \Delta p = \frac{1}{2}$.

Thus the wave packet is normalized and has a well-defined momentum distribution, with the properties of a momentum eigenstate while being relatively well localized in space for finite Δp . By using Gaussian wave packets, we can compute the S-matrix elements for the scattering process without encountering divergences, and we can ensure that the initial and final states are well-defined and physically meaningful.

Initial states. Sticking to the Gaussian wave packet example, we can write the initial states as the following superposition of momentum eigenstates:

$$\begin{aligned} |p_a\rangle \rightarrow |\phi_a\rangle &= \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \phi_a(\mathbf{k}) |\mathbf{k}\rangle, \\ |p_b\rangle \rightarrow |\phi_b\rangle &= \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \phi_b(\mathbf{k}) e^{-i\mathbf{k}\cdot\mathbf{b}} |\mathbf{k}\rangle, \end{aligned}$$

where we exploited the dependence of $\phi_b(\mathbf{k})$ upon $\mathbf{b}$ the impact parameter as

$$\mathbf{b} = \langle \phi_b | \hat{\mathbf{x}}^\perp | \phi_b \rangle,$$

with $\hat{\mathbf{x}}^\perp$ being the position operator in the plane perpendicular to the beam axis.

Outgoing states. For the outgoing states we can consider just the momentum eigenstates, since the scattering has already happened, thus we can write

$$|\phi_i\rangle = |p_i\rangle, \quad \forall i = 1, \dots, N.$$

Since the outgoing particles are detected at macroscopic distances, we can treat them as free particles and thus we can use momentum eigenstates to describe them: the scattering already happened, thus we can treat the outgoing particles as free particles.

Now we can compute the **S-matrix elements** for the scattering process using the initial and final states defined above. We can write the S-matrix elements for momentum eigenstates as

$$\langle f, \infty | i, -\infty \rangle = \langle f | \hat{S} | i \rangle = \langle p_1, \dots, p_N | \hat{S} | p_a, p_b \rangle,$$

where we are considering the initial state as a two particle state with momenta p_a and p_b , and the final state as a N particle state with momenta $p_1, \dots, p_N$. The S-matrix can be expressed as

$$\hat{S} = \mathbb{I} + i\hat{T},$$

where the identity term corresponds to the case where no scattering occurs, while the $i\hat{T}$ term corresponds to the case some interactions among the fields, i.e. the scattering occurs. Thus we can write¹

$$\langle p_1, \dots, p_N | i\hat{T} | p_a, p_b \rangle = (2\pi)^4 \delta^{(4)}(p_a + p_b - \sum_{i=1}^N p_i) i\mathcal{M},$$

where the delta function ensures the conservation of energy and momentum, and $\mathcal{M}$ is the **invariant matrix element** or **scattering amplitude** for the process, which encodes the dynamics of the interaction (we omitted its dependencies for simplicity, but it relies on the initial and final states, thus $\mathcal{M} = \mathcal{M}(p_a, p_b \rightarrow p_1, \dots, p_N)$). The invariant amplitude can be computed using Feynman diagrams and the rules of quantum field theory, and it is a key quantity in determining the cross section for the scattering process.

Thus we can now compute the **differential cross section** in the space of final configurations as in (2.3.4):

$$\frac{d\sigma}{df} = \int d^2\mathbf{b} \frac{dP}{df},$$

¹Note that we evaluate the transition amplitude here using exact momentum eigenstates $|p_a, p_b\rangle$ rather than the initial wave packets $|\phi_a, \phi_b\rangle$. This is done to formally define the invariant scattering amplitude $\mathcal{M}$ and cleanly factor out the energy-momentum conserving Dirac delta $\delta^{(4)}$. The physical wave packets enter the calculation in the subsequent step: exploiting the linearity of the $\hat{T}$ operator, the realistic transition amplitude is obtained by integrating these exact-momentum matrix elements over the momentum distributions $\phi_a(\mathbf{k})$ and $\phi_b(\mathbf{k})$ of the incoming wave packets.

where the differential probability of scattering in the space of final states is given by

$$\frac{dP}{df} = |\langle p_1, \dots, p_N, +\infty | i\hat{T} | \phi_a, \phi_b, -\infty \rangle|^2,$$

and the measure of the space of final states for N particles is normalized as²

$$df = \prod_{i=1}^N \frac{d^3\mathbf{p}_i}{(2\pi)^3} \frac{1}{2E_{p_i}}.$$

Now we can put everything back together and after some algebra (full computation in Appendix B) we can write the final expression for the differential cross section

TODO: write it.

$$\begin{aligned} d\sigma &= \frac{1}{\Phi} |\mathcal{M}(p_a, p_b \rightarrow p_1, \dots, p_N)|^2 d\Pi_N, \\ d\Pi_N &= \left[\prod_{i=1}^N \frac{d^3\mathbf{p}_i}{(2\pi)^3 2E_{p_i}} \right] (2\pi)^4 \delta^{(4)}(p_a + p_b - \sum_{i=1}^N p_i), \\ \Phi &= 2E_a 2E_b |v_a - v_b| = 4|E_b p_a^z - E_a p_b^z|, \end{aligned}$$

where:

- $d\Pi_N$ is the Lorentz-invariant phase space measure for N final state particles, which accounts for the density of final states and the conservation of energy and momentum through the delta function.
- Φ is the flux factor, which accounts for the incoming particles and their relative velocity, ensuring that the cross section is properly normalized with respect to the initial state of the scattering process.
- $\mathcal{M}$ is again the invariant matrix element or scattering amplitude, which encodes the dynamics of the interaction and can be computed using Feynman diagrams and the rules of quantum field theory.

Φ is the only non Lorentz invariant term: we have indeed

$$\Phi = 4|E_b p_a^z - E_a p_b^z|,$$

where z is the beam axis, thus it is not invariant. But we can recognize its invariance under:

1. boosts along the beam axis, since E_a and p_a^z transform in the same way, and E_b and p_b^z also transform in the same way, thus the combination $E_b p_a^z - E_a p_b^z$ is invariant under boosts along the beam axis.
2. rotations around the beam axis (in the perpendicular plane), since E_a , E_b , p_a^z and p_b^z are scalars under rotations around the beam axis, thus the combination $E_b p_a^z - E_a p_b^z$ is invariant under rotations around the beam axis.

²We can convince ourself with the resolution of the identity for 1 particle states, which reads

$$\mathbb{I} = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{2E_{\mathbf{k}}} |\mathbf{k}\rangle \langle \mathbf{k}|,$$

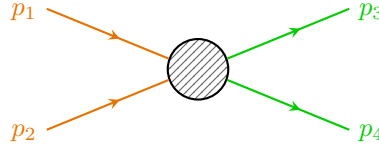
where $df = \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{2E_{\mathbf{k}}}$ is the measure of the space of final states for 1 particle.

Thus the flux factor Φ transforms as the inverse of an area in the plane perpendicular to the beam axis, behaving as a cross section (which is consistent with the interpretation of the cross section as an effective area that quantifies the likelihood of a scattering event occurring when a particle from the beam interacts with the target).

Even if we focus on the case of a beam of particles scattering on a target, the computed expression for the differential cross section remains symmetric for exchange of beam and target, thus it can be applied to any scattering process, regardless of the specific roles of the particles involved. We can also imagine to use it for collisions of two beams of particles, as in e^+e^- collisions in LEP, $p\bar{p}$ collisions in the Tevatron, or more recently pp collisions in the LHC.

We are basically performing a change of reference frame in order to reverse the problem: we can consider the target as the beam and the beam as the target, it all depends on the symmetries and the choice of the reference frame. This is a common technique in scattering theory, where we can choose a reference frame that simplifies the calculations (even center of mass, it depends) and makes it easier to analyze the scattering process. By considering different reference frames, we can gain insights into the dynamics of the interaction and compute cross sections more efficiently.

Example ($2 \rightarrow 2$ scattering in center of mass frame). We can consider a simple scattering process where two particles collide and produce two particles in the final state. In the center of mass frame, the initial momenta of the particles are equal and opposite, and we can analyze the scattering process using the invariant amplitude $\mathcal{M}$ and the phase space factor Φ to compute the differential cross section for this process.



The incoming particles have momenta p_1 and p_2 , and the outgoing particles have momenta p_3 and p_4 , and their spatial parts is conserved: $\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{p}_3 + \mathbf{p}_4 = 0$ (since we are in the center of mass frame). The energy of the particles is given by the energy momentum relation $E_i = \sqrt{\mathbf{p}_i^2 + m_i^2}$, where m_i is the mass of the particle. We can write

$$d\Pi_2 = \frac{d^3\mathbf{p}_3}{(2\pi)^3 2E_3} \frac{d^3\mathbf{p}_4}{(2\pi)^3 2E_4} (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4),$$

which, after integrating in $d^3\mathbf{p}_4$ using the delta function, gives $\mathbf{p}_4 = -\mathbf{p}_3$; using then polar coordinates for $d^3\mathbf{p}_3 = p^2 dp d\Omega$, we can write

$$E_3 = \sqrt{p^2 + m_3^2}, \quad E_4 = \sqrt{p^2 + m_4^2}, \quad p = |\mathbf{p}_3| = |\mathbf{p}_4|,$$

and thus

$$\int dp \delta(E_1 + E_2 - E_3 - E_4) = \left| \frac{p}{E_3} + \frac{p}{E_4} \right|^{-1} = \frac{E_3 E_4}{p(E_3 + E_4)} = \frac{E_3 E_4}{pE},$$

where the last energy E is the total energy of the system in the center of mass frame, which is given by $E = E_1 + E_2 = E_3 + E_4$. Thus we can write

$$d\Pi_2 = d\Omega \frac{p}{16\pi^2 E},$$

with a flux $\Phi = 2E_1 2E_2 |v_1 - v_2|$, with now

$$|v_1 - v_2| = \left| \frac{|\mathbf{p}_1|}{E_1} + \frac{|\mathbf{p}_2|}{E_2} \right| = p_{in} \frac{E}{E_1 E_2},$$

where p_{in} is the magnitude of the incoming momenta (which are equal in the center of mass frame) $p_{in} = |\mathbf{p}_1| = |\mathbf{p}_2|$. Thus we can write the final expression for the differential cross section as

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 E^2} \frac{p}{p_{in}} |\mathcal{M}|^2,$$

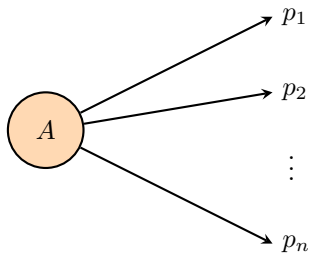
where we used the fact that $E_1 E_2 = \frac{E^2}{4}$ in the center of mass frame. This expression allows us to compute the *differential cross section for the $2 \rightarrow 2$ scattering process only in the center of mass frame*, given the invariant amplitude $\mathcal{M}$ and the kinematic variables of the process.

Remark. *Most of the time this problem is independent of the azimuthal angle ϕ , thus we can integrate over it and write $d\Omega = 2\pi d\cos\theta$, thus simplifying the expression for the differential cross section.*

Furthermore $d\sigma$ is invariant under boosts along the beam axis, but the solid angle $d\Omega$ is not, thus the expression for the differential cross section $\frac{d\sigma}{d\Omega}$ is not invariant.

2.3.3 | Decay Rates

We can similarly deal with **unstable particles**, which after a certain lifetime can decay into other lighter particles. Systems composed by unstable particles can be analyzed using the S-matrix formalism in a similar way to how we compute scattering amplitudes and cross sections for stable particles. By analyzing the transition amplitudes for the decay process, we can determine the decay rate and the branching ratios for different decay channels, which provide insights into the properties of the unstable particle and its interactions with other particles.



The **decay rate** Γ of an unstable particle can be computed as

$$\Gamma = \frac{\text{Number of decays per unit time}}{\text{Number of particles}},$$

and we can define the **lifetime** τ of the particle as the inverse of the decay rate:

$$\tau = \frac{1}{\Gamma}.$$

Decay rates and lifetimes are defined in the rest frame of the particle, where the particle is at rest and its energy is equal to its mass $E = m$ and its momentum is zero $\mathbf{p} = 0$.

The decay rate can be computed using the S-matrix element for the transition from the initial state (the unstable particle) to the final state (the decay products), and it is related to the **invariant amplitude** $\mathcal{M}$ for the decay process. We cannot directly apply the limit for $t_i \rightarrow -\infty$, since the particle is unstable and it would have decayed before, it cannot exist for an infinite amount of time. We therefore need to consider a finite time interval for the decay process, but we can make a simplification: if we assume the particle to have a lifetime way longer than the time scale of the interaction (always true for most practical cases), we can still apply the S-matrix formalism and

compute the decay rate using the invariant amplitude for the decay process, and we will have to take into account the instability of the particle only in one step of the calculation.³

Defining the **differential decay rate** as

$$\frac{d\Gamma}{df} = \frac{1}{T} dP, \quad (2.3.5)$$

where T is the time interval during which the decay process occurs $T = t_f - t_i$, and dP is the probability of decaying from the initial state to the final state within that time interval; we can compute the total decay rate by integrating over the phase space of the final states; let us first write the expression for the delta function in the limit of finite and large T :

$$\delta(0) = \int_{t_i}^{t_f} dt e^{i(E_i - E_f)t} \Big|_{t_f \rightarrow \infty, t_i \rightarrow -\infty},$$

which is the critical step in the calculation, since it allows us to regularize the delta function and make sense of the transition amplitude for the decay process; the complete calculation is shown in Appendix B. Having solved the subtleties of defining the asymptotic states for unstable particles, the final expression for the decay rate after integrating over the phase space of the final states is

$$d\Gamma = \frac{1}{2m} |\mathcal{M}_{p_i \rightarrow p_f}|^2 d\Pi_N, \quad (2.3.6)$$

where m is the mass of the unstable particle, $\mathcal{M}_{p_i \rightarrow p_f}$ is the invariant amplitude for the decay process, and $d\Pi_N$ is the phase space factor for the N -particle final state, which accounts for the kinematics of the decay products and ensures that energy and momentum are conserved in the decay process. It is very similar to the expression for the differential cross section, with some differences concerning the flux factor term $\Phi = 2E_a 2E_b |v_a - v_b|$:

- $2E_a \rightarrow 2m$ since we are in the rest frame of the unstable particle, thus $E_a = m$;
- $2E_b |v_a - v_b|$ is not present since we are not dealing with a scattering process, but rather with a decay process, thus there is no incoming particle b and no relative velocity to consider.

This expression allows us to compute the decay rate for an unstable particle given the invariant amplitude and the kinematic variables of the decay process.

Identical particles. When dealing with identical particles among the decay products, the previous formula are still valid, with the appropriate modifications to take into account the symmetrization (for bosons) or antisymmetrization (for fermions) of the wave function. This means that the scattering amplitude must be modified to include the appropriate symmetry factors, which can affect the calculation of cross sections and decay rates in order to consider the spin statistics theorem

$$|p_3, p_4\rangle = \pm |p_4, p_3\rangle.$$

This can be accomplished in two ways: one can restrict the integration over the phase space to a specific region that accounts for the indistinguishability of the particles (thus over all final inequivalent states), or one can include a symmetry factor in the calculation of the amplitude (such as the permutation number $\frac{1}{N!}$) to account for the number of identical particles in the final state. Both approaches ensure that the physical predictions are consistent with the principles of quantum mechanics and the statistics of identical particles.

³We will consider a finite time interval for the decay process and then take the limit of large time intervals to recover the usual expression for the decay rate.

TODO: check footnote.

TODO: write it.

2.4 | LSZ reduction and Green's functions

In this section we aim to show how the S-matrix elements can be computed from the Green's functions of the interacting theory. This is done by the LSZ reduction formula, which relates the S-matrix elements to the time-ordered correlation functions of the fields.

The Green's functions of the interacting theory are defined as the time-ordered correlation functions of the field operators in the interacting theory, evaluated in the vacuum state of the interacting theory:

$$\langle f | \hat{S} | i \rangle \rightarrow \langle \Omega | T \{ \hat{\phi}(x_1) \hat{\phi}(x_2) \cdots \hat{\phi}(x_n) \} | \Omega \rangle \quad (2.4.1)$$

where

- $\hat{\phi}$ is the field operator of the interacting theory,
- $|\Omega\rangle$ is the vacuum state of the interacting theory,

where we highlighted the difference among the vacuum state of the interacting theory and the free theory one: interactions can modify the structure of the vacuum and the Green's functions of the interacting theory encode all the information about its dynamics; they are the building blocks for computing physical observables such as scattering amplitudes and cross sections.

We now start from the structure analysis of the two point functions $\langle \Omega | \hat{T} \{ \hat{\phi}(x) \hat{\phi}(y) \} | \Omega \rangle$ in an interacting theory, investigating its physical content and how it relates to the S-matrix elements $\langle f | \hat{S} | i \rangle$, finally relating it to the LSZ reduction formula for connecting the scattering matrix elements to the n-point Green's functions. We will start from a **real scalar theory** before generalizing to more complicated theories.

2.4.1 | Two Point Functions

To analyze the structure of the two point function, we start from the definition of the time-ordered two point function in the interacting theory:

$$\langle \Omega | T \{ \hat{\phi}(x) \hat{\phi}(y) \} | \Omega \rangle,$$

the idea is now to insert a complete set of physical states $\langle \lambda |^4$ as a resolution of the identity $\mathbb{I} = \sum_{\lambda} |\lambda\rangle \langle \lambda|$ between the two field operators; but we have to be more precise, since the vacuum state $|\Omega\rangle$ is a physical state of the theory, we have to separate it from the other states in the spectrum of the theory, so that we can write the resolution of the identity as

$$\mathbb{I} = |\Omega\rangle \langle \Omega| + \sum_{\lambda \neq \Omega} |\lambda\rangle \langle \lambda|,$$

where $|\lambda\rangle$ are seen as eigenstates of the momentum operator $\hat{\mathbf{P}}$ plus other quantum numbers, so that we can write them as $|\lambda_{\mathbf{p}}\rangle$ to make the momentum dependence explicit. They are normalized according to the relativistic normalization convention, so that we have

$$\langle \lambda_{\mathbf{p}} | \lambda'_{\mathbf{p}'} \rangle = (2\pi)^3 2E_{\mathbf{p}}(\lambda) \delta^{(3)}(\mathbf{p} - \mathbf{p}') \delta_{\lambda, \lambda'},$$

⁴Here we can interpret λ as the set of quantum numbers characterizing the different states of our theory.

TODO: Add reference to Peskin 7.1

where $E_{\mathbf{p}}(\lambda) = \sqrt{\mathbf{p}^2 + m_\lambda^2}$ is the energy of the state $|\lambda_{\mathbf{p}}\rangle$ with momentum $\mathbf{p}$ and mass m_λ . The sum over λ includes all the physical states of the theory, such as the one particle states, the two particle states, the bound states, etc.

We cited the mass m_λ of the state $|\lambda_{\mathbf{p}}\rangle$ as a label for the state, but it is important to note that the mass m_λ is not a free parameter, but it is determined by the dynamics of the theory and it is related to the eigenvalue of the momentum operator $\hat{P}^\mu$ on the state $|\lambda_{\mathbf{p}}\rangle$ by the relativistic dispersion relation:

$$\hat{P}^\mu |\lambda_{\mathbf{p}}\rangle = p^\mu |\lambda_{\mathbf{p}}\rangle, \quad p^2 = p_\mu p^\mu = m_\lambda^2,$$

so that the mass m_λ is determined by the eigenvalue of $\hat{P}^2$ on the state $|\lambda_{\mathbf{p}}\rangle$:

$$\hat{P}^2 |\lambda_{\mathbf{p}}\rangle = m_\lambda^2 |\lambda_{\mathbf{p}}\rangle.$$

Putting everything together, we can write the resolution of the identity as

$$\mathbb{I} = |\Omega\rangle \langle\Omega| + \sum_{\lambda \neq \Omega} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}|,$$

with the factor of $1/(2E_{\mathbf{p}}(\lambda))$ coming from the relativistic normalization of the states $|\lambda_{\mathbf{p}}\rangle$.

If we now assume $x^0 > y^0$, the time-ordering operator T does not change the order of the fields, and we can neglect it in order to write the two point function as:

$$\begin{aligned} \langle\Omega| \hat{\phi}(x) \hat{\phi}(y) |\Omega\rangle &= \langle\Omega| \hat{\phi}(x) \left(|\Omega\rangle \langle\Omega| + \sum_{\lambda \neq \Omega} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| \right) \hat{\phi}(y) |\Omega\rangle \\ &= \langle\Omega| \hat{\phi}(x) |\Omega\rangle \langle\Omega| \hat{\phi}(y) |\Omega\rangle + \sum_{\lambda \neq \Omega} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} \langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| \hat{\phi}(y) |\Omega\rangle, \end{aligned}$$

and if we now relate $\hat{\phi}(x)$ to $\hat{\phi}(0)$ by a spacetime translation, using the unitary operator $\hat{U}_{\mathbf{p}} = e^{i\hat{P}\cdot x}$ that implements the translation, so that

$$\hat{U}_{\mathbf{p}} = e^{i\hat{P}\cdot x} \implies \hat{\phi}(x) = e^{i\hat{P}\cdot x} \hat{\phi}(0) e^{-i\hat{P}\cdot x} = \hat{U}_{\mathbf{p}} \hat{\phi}(0) \hat{U}_{\mathbf{p}}^\dagger,$$

we can compute the terms by assuming that the vacuum is *translationally invariant*, and since we said that $|\lambda_{\mathbf{p}}\rangle$ are eigenstates of the momentum operator $\hat{P}$ with eigenvalue p , we can write:

$$\begin{aligned} \langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle &= \langle\Omega| e^{i\hat{P}\cdot x} \hat{\phi}(0) e^{-i\hat{P}\cdot x} |\lambda_{\mathbf{p}}\rangle \\ &= e^{-ip\cdot x} \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{p}}\rangle, \end{aligned}$$

from vacuum invariance and momentum eigenstate properties. We can now assume a Lorentz transformation Λ and the corresponding unitary operator $\hat{U}(\Lambda)$ to exist such that $\Lambda p = (m_\lambda, \mathbf{0})$ and $\hat{U}(\Lambda) |\lambda_{\mathbf{p}}\rangle = |\lambda_{\mathbf{0}}\rangle$. Hence

$$\langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{p}}\rangle = \left(\langle\Omega| \hat{U}(\Lambda)^\dagger \right) \left(\hat{U}(\Lambda) \hat{\phi}(0) \hat{U}(\Lambda)^\dagger \right) \left(\hat{U}(\Lambda) |\lambda_{\mathbf{p}}\rangle \right) = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle,$$

since the vacuum is invariant under Lorentz transformations, and the field operator transforms trivially: $\hat{\phi}(\Lambda 0) = \hat{\phi}(0)$. Thus we can write

$$\langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle e^{-ip\cdot x} = e^{-ip\cdot x} F(\lambda),$$

where we have defined the function $F(\lambda) = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle$ which encodes the information about the coupling of the field operator to the state $|\lambda\rangle$. In the particular case of $|\lambda_{\mathbf{p}}\rangle = |\Omega\rangle$, we get

$$\langle\Omega| \hat{\phi}(x) |\Omega\rangle = v = \text{constant},$$

(?) why the physical states can be transformed to the rest frame?

which is a consequence of the translational invariance of the vacuum state and represents the vacuum expectation value of the field operator. We can then always redefine the field operator by subtracting the vacuum expectation value

$$\phi(x) = v + \tilde{\phi}(x), \quad \langle \Omega | \tilde{\phi}(x) | \Omega \rangle = 0,$$

since the vacuum expectation value of the field operator does not affect the dynamics of the theory, and it can be absorbed into a redefinition of the field operator. This means that we can always choose a field operator such that its vacuum expectation value is zero, which simplifies the analysis of the two point function and the LSZ reduction formula. Thus the two point (after the redefinition of the field operator) function can be written as

$$\langle \Omega | \hat{\phi}(x) \hat{\phi}(y) | \Omega \rangle = \sum_{\lambda} |F(\lambda)|^2 \int \frac{d^3 \mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} e^{-ip \cdot (x-y)},$$

valid for $x^0 > y^0$ and where the last term

$$D(x, y, m_{\lambda}) = \int \frac{d^3 \mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} e^{-ip \cdot (x-y)} = \langle 0 | \hat{\phi}_{0,\lambda}(x) \hat{\phi}_{0,\lambda}(y) | 0 \rangle, \quad (2.4.2)$$

is the **free propagator** of a particle with mass m_{λ} , with $\hat{\phi}_{0,\lambda}$ the free field operator of a particle with mass m_{λ} : it is the amplitude for a free scalar particle to propagate from point y to point x . We can now reintroduce the time-ordering to write the two point function in a more general form

$$\begin{aligned} \langle \Omega | T \{ \hat{\phi}(x) \hat{\phi}(y) \} | \Omega \rangle &= \Theta(x^0 - y^0) \langle \Omega | \hat{\phi}(x) \hat{\phi}(y) | \Omega \rangle + \Theta(y^0 - x^0) \langle \Omega | \hat{\phi}(y) \hat{\phi}(x) | \Omega \rangle \\ &= \Theta(x^0 - y^0) \sum_{\lambda} |F(\lambda)|^2 D(x, y, m_{\lambda}) + \Theta(y^0 - x^0) \sum_{\lambda} |F(\lambda)|^2 D(y, x, m_{\lambda}). \end{aligned}$$

In this last expression we can recognize the **Feynman propagator** of a particle with mass m_{λ} :

$$\begin{aligned} D_F(x - y, m_{\lambda}) &= \Theta(x^0 - y^0) D(x, y, m_{\lambda}) + \Theta(y^0 - x^0) D(y, x, m_{\lambda}) \\ &= \langle 0 | \hat{T} \{ \hat{\phi}_{0,\lambda}(x) \hat{\phi}_{0,\lambda}(y) \} | 0 \rangle = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m_{\lambda}^2 + i\epsilon} e^{-ip \cdot (x-y)}, \end{aligned} \quad (2.4.3)$$

which is the amplitude for a free scalar particle to propagate from point y to point x in a time-ordered manner, and it is given by the Fourier transform of the free propagator with the appropriate $i\epsilon = i0^+$ prescription to ensure causality.

With this last definition, we can now write the final expression, the **Källén-Lehmann spectral representation** of the two point function, as:

$$\langle \Omega | T \{ \hat{\phi}(x) \hat{\phi}(y) \} | \Omega \rangle = \int \frac{dM}{2\pi} g(M^2) D_F(x, y, M), \quad (2.4.4)$$

where we have defined the **spectral density** function $g(M^2)$ as

$$g(M^2) = \sum_{\lambda} (2\pi) |F(\lambda)|^2 \delta(M^2 - m_{\lambda}^2). \quad (2.4.5)$$

The spectral density function encodes the information about the spectrum of the theory,⁵ and it is a positive function since it is defined as a sum of positive terms. The Källén-Lehmann representation of the two point function shows that the two point function can be written as a superposition of free propagators with different masses, weighted by the spectral density function.

⁵The spectral density function is a measure of the number of states with a given mass: the spectrum of the theory is the set of all possible masses of physical particles.

This spectral representation is very useful for analyzing the properties of the theory, and it will be crucial for deriving the LSZ reduction formula in the next section: it will allow us to relate the S-matrix elements to the poles of the two point function (which correspond to the physical particles in the spectrum of the theory) and to the residues of these poles (which correspond to the field strength renormalization constants of these particles).

Let us make some observations about the Källén-Lehmann representation to understand better the look and physical meaning of the spectral density function $g(M^2)$:

- If $M^2 = m^2$ with m the physical mass of a **one particle state** (i.e. m^2 eigenvalue of $\hat{P}^2$), then the spectral density function has a delta function contribution

$$g(M^2) \Big|_{M^2=m^2} = Z(2\pi)\delta(M^2 - m^2),$$

with $Z = |F(\lambda = 1\text{-part. state})|^2$ which represents the *field strength renormalization constant* of the one particle state. This means that the two point function has a pole at $p^2 = m^2$ with residue Z , which is a consequence of the presence of a one particle state in the spectrum of the theory.

- For a **two particle bound state** we have two cases:

- If $M^2 \leq (2m)^2$ then the spectral density function has a delta function contribution

$$g(M^2) \Big|_{M^2 \leq (2m)^2} = \sum_{\lambda \in 2\text{-part. states}} Z_B(2\pi)\delta(M^2 - m_\lambda^2),$$

with $Z_B = |F(\lambda = 2\text{-part. state})|^2$ which represents the *field strength renormalization constant* of the two particle bound state. This means that the two point function has a pole at $p^2 = M_B^2$ with residue Z_B , which is a consequence of the presence of a two particle bound state in the spectrum of the theory.

- If $M^2 > (2m)^2$ then the spectral density function has a continuous contribution given by the two particle continuum states, which represent the scattering states of two particles with mass m

$$g(M^2) = Z_C \sum_{\lambda \in 2\text{-part. cont. states}} (2\pi)\delta(M^2 - m_\lambda^2),$$

with $Z_C = |F(\lambda = 2\text{-part. cont. state})|^2$ which represents the *field strength renormalization constant* of the two particle continuum states. This means that the two point function has a branch cut starting at $p^2 = (2m)^2$, which is a consequence of the presence of the two particle continuum states in the spectrum of the theory.

- The spectral density function $g(M^2)$ is a positive function of the **bare mass** M , since it is defined as a sum of positive terms. This means that the two point function is a positive function, which is a consequence of the unitarity of the theory.
- The physical mass m^2 is the eigenvalue of $\hat{P}^2$, so that it represents the mass of the one particle state in the spectrum of the theory. The physical mass is different from the bare mass M^2 which appears in the Lagrangian of the theory, and it is related to the bare mass by a renormalization procedure.

Thus we can now understand a typical look of the spectral density function $g(M^2)$ for a theory with a one particle state, some two particle bound states and a two particle continuum states:

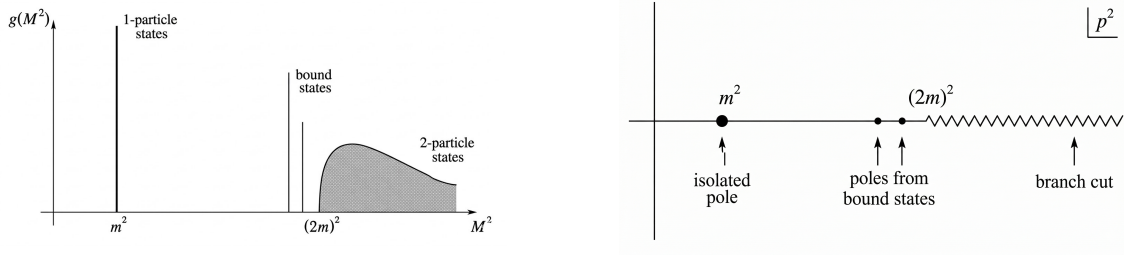


Figure 2.4: On the left image we can see the spectral density function for a theory with a one particle state and a two particle bound and continuum states, with the delta function contributions and the continuous part. On the right image we can see the analytic structure in the complex p^2 -plane of the two point function Fourier transform, with the pole at $p^2 = m^2$ corresponding to the one particle state, the poles at $p^2 = M_B^2 \leq (2m)^2$ corresponding to the two particle bound states, and the branch cut starting at $p^2 = (2m)^2$ corresponding to the two particle continuum states.

We can now study again the two point function in momentum space, by performing a Fourier transform of the Källén-Lehmann representation:

$$\begin{aligned} \int d^4x e^{ip \cdot x} \langle \Omega | T \{ \hat{\phi}(x) \hat{\phi}(0) \} | \Omega \rangle &= \int d^4x e^{ip \cdot x} \int \frac{dM}{2\pi} g(M^2) D_F(x, 0, M) \\ &= \frac{iZ}{p^2 - m^2 + i0^+} + \int_{(2m)^2}^{\infty} \frac{dM^2}{2\pi} \frac{ig(M^2)}{p^2 - M^2 + i\epsilon}, \end{aligned}$$

where we have divided the integral over M^2 into the contribution from the one particle state, which gives a pole at $p^2 = m^2$ with residue iZ , and the contribution from the two particle states, which gives a branch cut starting at $p^2 = (2m)^2$ (these are regular terms for $p^2 \sim m^2$). The contribution from the two particle bound states, if present, would give poles at $p^2 = M_B^2 \leq (2m)^2$ with residue iZ_B (as shown in the above figure), but we have not explicitly written them in the last expression for simplicity. To sum up, in momentum space the two point function has the following analytic structure in the complex p^2 -plane:

1. A branch cut starting at $p^2 = (2m)^2$ due to the presence of the two particle continuum states in the spectrum of the theory, which represent the scattering states of two particles with mass m .
2. A pole at $p^2 = m^2$ with residue iZ , which is a consequence of the presence of a one particle state in the spectrum of the theory.
3. Poles at $p^2 \leq (2m)^2$ for each two particle bound state in the spectrum of the theory, with residue iZ_B , which is a consequence of the presence of the two particle bound states in the spectrum of the theory.

For a free theory we would have only the one particle state in the spectrum of the theory, so that the Källén-Lehmann representation of the two point function would reduce to

$$\int d^4x e^{ip \cdot x} \langle 0 | T \{ \hat{\phi}_0(x) \hat{\phi}_0(0) \} | 0 \rangle = \frac{i}{p^2 - m^2 + i0^+},$$

with $Z = 1$ since the field operator creates a one particle state with unit probability, and the physical mass m^2 coincides with the bare mass M^2 since there are no interactions in the theory.

For an interacting theory we will have corrections to the physical mass and to the field strength renormalization constant, so that we can write

$$\begin{cases} m = m_\lambda + O(g), \\ Z = 1 + O(g), \end{cases}$$

where g is the coupling constant of the theory, so that the physical mass m^2 and the field strength renormalization constant Z receive corrections from the interactions in the theory.

Example (Dirac Field). In the end we will give rules for the generalization of the LSZ reduction formula to more complicated theories, such as theories with fermions and gauge fields, but let us first analyze the structure of the two point function for a Dirac field. If we compute these last results for a Dirac field, we would find

$$\int d^4x e^{ip \cdot x} \langle \Omega | \hat{T} \{ \hat{\psi}(x) \hat{\bar{\psi}}(0) \} | \Omega \rangle = \frac{iZ(\not{p} + m)}{p^2 - m^2 + i\epsilon},$$

such that

$$\langle \Omega | \hat{\psi}(0) | \lambda_{\mathbf{p},s} \rangle \Big|_{\lambda=1\text{-part}} = u_s(\mathbf{p}) F(\lambda=1\text{-part}) = u_s(\mathbf{p}) \sqrt{Z},$$

where $u_s(\mathbf{p})$ is the Dirac spinor for a particle with momentum $\mathbf{p}$ and spin s , with

$$(\not{p} + m) = \sum_s u_s(\mathbf{p}) \bar{u}_s(\mathbf{p}),$$

which is the completeness relation for the Dirac spinors. The spectral density function for the Dirac field would have a delta function contribution at $M^2 = m^2$ with residue Z , which corresponds to the one particle state in the spectrum of the theory, and a branch cut starting at $M^2 = (2m)^2$ due to the presence of the two particle continuum states in the spectrum of the theory, which represent the scattering states of two particles with mass m . We would not have two particle bound states in the spectrum of the theory, since the Dirac field does not have self-interactions, so that the spectral density function would not have delta function contributions at $M^2 \leq (2m)^2$ corresponding to two particle bound states.

Example (Unstable Particle State). If we are considering unstable particles, i.e particles with a decay rate $\Gamma \neq 0$, then the last results are not valid anymore, since we adopted eigenvalues of the momentum operator $\hat{P}^\mu$, and in particular of $\hat{H}$, which cannot be defined for unstable particles:

$$\hat{H} |\psi\rangle = E |\psi\rangle, \hat{S}(t, t_0) |\psi\rangle = e^{i\hat{H}(t-t_0)} |\psi\rangle = e^{iE(t-t_0)} |\psi\rangle,$$

since an eigenstate by itself can only get a phase and does not decay, while an unstable particle state has to decay in order to be unstable.

Thus a rigorous definition of the physical mass for an unstable particle is rather non-trivial; however the decay rate Γ of the unstable particle state is usually very small compared to its lifetime, so that we can consider those as approximated eigenstates of the Hamiltonian.

We have to take into account the next facts:

- If $\Gamma \neq 0$ the pole of the two point function is not anymore on the real axis, but it is shifted in the complex plane, so that it acquires an imaginary part proportional to the decay rate Γ of the unstable particle state.
- If $\Gamma \ll m$, we may still define the physical mass m of the unstable particle state as the real part of the location of the pole of the two point function, since the imaginary part is

(?) Is this last statement true?

small compared to the real part, and it represents a small correction to the physical mass of the unstable particle state.

Thus in the end we can define the physical mass m of the unstable particle state as

$$m^2 = \text{Re} [\text{propagator pole}],$$

which gives us a well-defined physical mass for the unstable particle state, and we can write the two point function in momentum space as

$$\int d^4x e^{ip \cdot x} \langle \Omega | T \{ \hat{\phi}(x) \hat{\phi}(0) \} | \Omega \rangle = \frac{iZ}{p^2 - m^2 + im\Gamma} + \Lambda(p^2),$$

where $\Lambda(p^2)$ is a regular function for $p^2 \sim m^2$ which encodes the regular contributions from the other states in the spectrum of the theory.

2.4.2 | LSZ Reduction Formula

Let us start from the definition of the S-matrix elements in terms of the asymptotic in and out states:

$$\langle f | \hat{S} | i \rangle,$$

where $|i\rangle$ and $|f\rangle$ are the in and out states of the theory, which are defined as the asymptotic states of the theory at $t \rightarrow -\infty$ and $t \rightarrow +\infty$ respectively. Asymptotically, the interacting theory reduces to a free theory, so that the in and out states can be approximated by the free particle states of the free theory. This means that the field operators of the interacting theory can be approximated by the field operators of the free theory in this limit, in which they act as creation and annihilation operators for the free particle states (or 1-particle):

$$\hat{\phi}(x) \Big|_{t \rightarrow \pm\infty} \sim \sqrt{Z} \int \frac{d^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} (\hat{a}_{\mathbf{p}} e^{-ip \cdot x} + \hat{a}_{\mathbf{p}}^\dagger e^{ip \cdot x}),$$

where Z is the field strength renormalization constant of the one particle state in the spectrum of the theory, and $\hat{a}_{\mathbf{p}}$ and $\hat{a}_{\mathbf{p}}^\dagger$ are the annihilation and creation operators for the free particle states of the free theory. The energy appearing in the denominator is the physical energy of the one particle state, which is related to the physical mass m by the relativistic dispersion relation $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2} = p^0$.

Remark. Note that the annihilation and creation operators $\hat{a}_{\mathbf{p}}$ and $\hat{a}_{\mathbf{p}}^\dagger$ are time independent in a free theory. We will assume that they are also time independent in the interacting theory, when considering the asymptotic limit $t \rightarrow \pm\infty$, even if in general

$$\hat{a}_{\mathbf{p}}(t \rightarrow \infty) \neq \hat{a}_{\mathbf{p}}(t \rightarrow -\infty),$$

since the interactions can change the number of particles in the theory, so that the annihilation and creation operators at $t \rightarrow +\infty$ can be different from those at $t \rightarrow -\infty$ (the free-like field configurations may be different after the scattering process).

If we consider a scattering process with 2 incoming particles and n outgoing particles, we can write the in and out states as

$$\begin{aligned} |i\rangle &= \sqrt{2E_a} \sqrt{2E_b} \hat{a}_{\mathbf{p}_a}^\dagger \hat{a}_{\mathbf{p}_b}^\dagger |\Omega\rangle, \\ |f\rangle &= \sqrt{2E_1} \sqrt{2E_2} \cdots \sqrt{2E_n} \hat{a}_{\mathbf{p}_1}^\dagger \hat{a}_{\mathbf{p}_2}^\dagger \cdots \hat{a}_{\mathbf{p}_n}^\dagger |\Omega\rangle, \end{aligned}$$

TODO: Add reference to Schwartz 6.1

where we have properly normalized the states by the factors of $\sqrt{2E}$ according to the relativistic normalization convention. It is now time to compute the S-matrix element $\langle f | \hat{S} | i \rangle$ by inserting the expressions for the in and out states and recalling that $\hat{S} = \mathbb{I} + i\hat{T}$, such that we can write

$$\begin{aligned} \langle f | \hat{S} | i \rangle &= \langle f | \mathbb{I} + i\hat{T} | i \rangle = \langle f | i\hat{T} | i \rangle = \sqrt{2E_a} \sqrt{2E_b} \sqrt{2E_1} \sqrt{2E_2} \cdots \sqrt{2E_n} \times \\ &\times \langle \Omega | \hat{a}_{\mathbf{p}_1}(+\infty) \hat{a}_{\mathbf{p}_2}(+\infty) \cdots \hat{a}_{\mathbf{p}_n}(+\infty) \left(i\hat{T} \right) \hat{a}_{\mathbf{p}_a}^\dagger(-\infty) \hat{a}_{\mathbf{p}_b}^\dagger(-\infty) | \Omega \rangle. \end{aligned}$$

The idea is now to rewrite everything in terms of Green's functions of the theory, so let us first prove the following identity:

$$\begin{aligned} &\frac{i}{\sqrt{Z}} \int d^4x e^{ip \cdot x} (\partial_x^2 + m^2) \hat{T} \{ \hat{\phi}(x) \hat{O}(y_1, \dots, y_n) \} \\ &= \sqrt{2E_{\mathbf{p}}} \left(\hat{a}_{\mathbf{p}}(+\infty) \hat{T} \hat{O}(y_1, \dots, y_n) - \hat{T} \hat{O}(y_1, \dots, y_n) \hat{a}_{\mathbf{p}}(-\infty) \right), \end{aligned}$$

where $\hat{O}(y_1, \dots, y_n)$ is an arbitrary product of operators of the theory, and we have used the fact that the field operator $\hat{\phi}(x)$ can be approximated by the free field operator in the asymptotic limit.

In order to prove this identity, we now define $\hat{O}(x) = \hat{T} \{ \hat{\phi}(x) \hat{O}(y_1, \dots, y_n) \}$ as the time ordered product of the field operator and the arbitrary product of operators, so that we can write

$$\frac{i}{\sqrt{Z}} \int d^4x e^{ip \cdot x} (\partial_x^2 + m^2) \hat{O}(x) = \frac{i}{\sqrt{Z}} \int d^4x e^{ip \cdot x} (\partial_t^2 - \partial_i^2 + m^2) \hat{O}(x),$$

where we have explicitly written the derivatives in terms of the time and spatial derivatives. We can now integrate by parts the spatial derivatives, so that it changes sign and start acting on the left $\vec{\partial}_i \rightarrow \overleftarrow{\partial}_i$,⁶ getting to

$$\frac{i}{\sqrt{Z}} \int d^4x e^{ip \cdot x} (\partial_t^2 + \mathbf{p}^2 + m^2) \hat{O}(x),$$

where we can recognize the relativistic dispersion relation $E_{\mathbf{p}}^2 = \mathbf{p}^2 + m^2$ for a particle with mass m and momentum $\mathbf{p}$, so that we can write

$$\frac{1}{\sqrt{Z}} \int d^4x i e^{ip \cdot x} (\partial_t^2 + E_{\mathbf{p}}^2) \hat{O}(x) = \frac{1}{\sqrt{Z}} \int d^4x \partial_t \left(e^{ip \cdot x} (i\partial_t + E_{\mathbf{p}}) \hat{O}(x) \right).$$

Now we can split the integral in a spatial and a time integral, so that

$$\frac{1}{\sqrt{Z}} \int dt \partial_t e^{iE_{\mathbf{p}} t} \int d^3\mathbf{x} e^{-i\mathbf{p} \cdot \mathbf{x}} (i\partial_t + E_{\mathbf{p}}) \hat{O}(x),$$

which in this form is a total time derivative (the first integral), so that we can evaluate it at the boundaries $t = \pm\infty$, where we know the field operators to behave as free field operators, and the ladder operators to be time independent; so we can now work on the second integral and substitute the expression for $\hat{O}(x)$ and the asymptotic behavior of the field operator, getting to

$$\int \frac{d^3\mathbf{k}}{(2\pi)^3} \int d^3\mathbf{x} \left[\frac{E_{\mathbf{p}} + E_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}}} e^{-iE_{\mathbf{k}} t} e^{-i(\mathbf{p}-\mathbf{k}) \cdot \mathbf{x}} \hat{T} \{ \hat{a}_{\mathbf{k}} \hat{O} \} + \frac{E_{\mathbf{p}} - E_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}}} e^{iE_{\mathbf{k}} t} e^{i(\mathbf{p}-\mathbf{k}) \cdot \mathbf{x}} \hat{T} \{ \hat{a}_{\mathbf{k}}^\dagger \hat{O} \} \right],$$

in which a spatial integration gives us a delta function $\delta^{(3)}(\mathbf{p} - \mathbf{k})$ which allows us to perform the integral over $\mathbf{k}$, getting to

$$\begin{aligned} \frac{1}{\sqrt{Z}} \int d^4x i e^{ip \cdot x} (\partial_t^2 + E_{\mathbf{p}}^2) \hat{O}(x) &= \int dt \partial_t \left[e^{iE_{\mathbf{p}} t} \sqrt{2E_{\mathbf{p}}} \left(e^{-iE_{\mathbf{p}} t} \hat{T} \{ \hat{a}_{\mathbf{p}}(t) \hat{O} \} \right) \right] \\ &= \sqrt{2E_{\mathbf{p}}} \left[\hat{a}_{\mathbf{p}}(+\infty) \hat{T} \hat{O}(y_1, \dots, y_n) - \hat{T} \hat{O}(y_1, \dots, y_n) \hat{a}_{\mathbf{p}}(-\infty) \right], \end{aligned}$$

⁶We are assuming that the fields vanish at spatial infinity, so that the boundary terms from the integration by parts vanish: then the spatial derivatives will change sign and start acting on the left.

yielding the desired result. We can obtain a similar relation holding for its hermitian conjugate, which is

$$\frac{1}{\sqrt{Z}} \int d^4x (-i) e^{-ip \cdot x} (\partial_x^2 + m^2) \hat{O}(x) = \sqrt{2E_{\mathbf{p}}} \left[\hat{a}_{\mathbf{p}}^\dagger(+\infty) \hat{T} \hat{O}(y_1, \dots, y_n) - \hat{T} \hat{O}(y_1, \dots, y_n) \hat{a}_{\mathbf{p}}^\dagger(-\infty) \right].$$

Now we can apply the derived result to the expression for the S-matrix element $\langle f | \hat{S} | i \rangle$ that we wrote above, so that we can write

$$\langle f | \hat{S} | i \rangle = \sqrt{2E_a 2E_b 2E_1 2E_2 \dots 2E_n} \langle \Omega | \hat{T} \hat{a}_{\mathbf{p}_1}(+\infty) \hat{a}_{\mathbf{p}_2}(+\infty) \dots \hat{a}_{\mathbf{p}_n}(+\infty) (i\hat{T}) \hat{a}_{\mathbf{p}_a}^\dagger(-\infty) \hat{a}_{\mathbf{p}_b}^\dagger(-\infty) | \Omega \rangle,$$

where we inserted a time ordering operator T to use the derived result (the time ordering does not change the result since the ladder operators are already time-ordered). We can now apply the derived result for each of the ladder operators in the above expression: while we consider $\hat{a}_{\mathbf{p}_1}$ all the other ladder operators can be grouped in the arbitrary product of operators $\hat{O}$ in the derived result, so that we can apply it to each of the ladder operators in the above expression, getting to the famous **LSZ reduction formula**, which is essentially

$$\begin{aligned} \langle f | \hat{S} | i \rangle &= \left[\frac{i}{\sqrt{Z}} \int d^4x e^{ip_a \cdot x_a} (\partial^2 + m^2) \phi(x) \right] \left[\frac{i}{\sqrt{Z}} \int d^4y e^{ip_b \cdot x_b} (\partial^2 + m^2) \phi(y) \right] \\ &\times \left[-\frac{i}{\sqrt{Z}} \int d^4x_1 e^{-ip_1 \cdot x_1} (\partial^2 + m^2) \phi(x_1) \right] \dots \left[-\frac{i}{\sqrt{Z}} \int d^4x_n e^{-ip_n \cdot x_n} (\partial^2 + m^2) \phi(x_n) \right] \\ &\times \langle \Omega | T \{ \phi(x) \phi(y) \dots \phi(x_1) \phi(x_2) \dots \phi(x_n) \} | \Omega \rangle, \end{aligned} \quad (2.4.6)$$

which is a way to relate the S-matrix elements to the time-ordered correlation functions of the fields, which are the Green's functions of the theory. The LSZ reduction formula is a powerful tool for computing the S-matrix elements from the Green's functions, and it is widely used in quantum field theory to compute scattering amplitudes and cross sections for various processes.

The LSZ reduction formula can be interpreted as a way to amputate the external legs of the Green's functions, which correspond to the free propagators of the theory, and to put them on shell by applying the operator $(\partial^2 + m^2)$ to them, which gives zero when acting on the free propagators. The factors of $1/\sqrt{Z}$ in the formula are necessary to cancel the field strength renormalization constants that appear in the residues of the poles of the Green's functions, so that we can obtain finite results for the S-matrix elements.

Substantially, it is a bunch of Fourier transforms of the field operators, where the only difference is in the phase signs: incoming particles have a negative phase sign, while outgoing particles have a positive phase sign.

If we rewrite it in the momentum space, integrating by parts the derivatives so that they act on the exponentials on their left (which corresponds to the substitution $i(\partial_i^2 + m^2) \rightarrow -i(p_i^2 - m^2)$) we can write then the LSZ reduction formula as

$$\langle f | \hat{S} | i \rangle = \prod_{i=a,b,1,\dots,n} \left[\frac{i}{\sqrt{Z}} \int d^4x_i e^{\pm ip_i \cdot x_i} (\partial^2 + m^2) \right] \langle \Omega | T \{ \phi(x_a) \phi(x_b) \dots \phi(x_1) \phi(x_2) \dots \phi(x_n) \} | \Omega \rangle,$$

TODO: check formula.

We have assumed asymptotic particles as the final state, with on shell momenta $p_i^2 = m^2$, so that the factors of $(\partial^2 + m^2)$ in the LSZ reduction formula will give zero when acting on the external legs of the Green's functions, which are the free propagators of the theory. This means that the only contribution to the S-matrix elements will come from the poles of the Green's functions, which correspond to the physical particles in the spectrum of the theory. The residues of these poles will

give the field strength renormalization constants Z for each particle, which will cancel with the factors of $1/\sqrt{Z}$ in the LSZ reduction formula, giving a finite result for the S-matrix elements.

2.4.3 | Generalization to Other Fields

If we repeat the same analysis for different types of fields, such as Dirac fields or gauge fields, we will find similar results for the two point functions and the LSZ reduction formula, with some modifications due to the different spin and statistics of the fields:

- For a **complex scalar field**, we would have two types of particles, the particle and the antiparticle, with different creation and annihilation operators, and the LSZ reduction formula would involve both types of operators for the in and out states:
 - $\phi^\dagger$ for incoming particles and outgoing antiparticles, with a positive phase sign in the Fourier transform.
 - ϕ for outgoing particles and incoming antiparticles, with a negative phase sign in the Fourier transform.

This would lead to a more complicated LSZ reduction formula, but the general structure would be the same as for the real scalar field, and with this prescription we could also compute the results for Dirac theory ($\phi \rightarrow \psi$ and $\phi^\dagger \rightarrow \bar{\psi}$).

- If we are considering a theory of several fields, then the Green function will contain all the different fields along with the different masses of the quanta, and the LSZ reduction formula will involve the corresponding field operators for each type of particle in the in and out states, with the appropriate phase signs in the Fourier transforms.
- When considering particle with different spins, we have to take into account the different transformation properties of the fields under Lorentz transformations, which will affect the structure of the two point functions and the LSZ reduction formula. For example, for a Dirac field we would have the following field operators for the in and out states:
 - $\bar{u}_{s,\alpha}$ for outgoing particles;
 - $u_{s,\alpha}$ for incoming particles;
 - $v_{s,\alpha}$ for outgoing antiparticles;
 - $\bar{v}_{s,\alpha}$ for incoming antiparticles.

If we instead consider a spin-1 gauge field, we would have the following field operators for the in and out states:

- $\epsilon_\mu(p, \lambda)$ for outgoing particles;
- $\epsilon_\mu^*(p, \lambda)$ for incoming particles.

The spinorial/vectorial indices (α, μ etc.) of the field operators will be contracted with the corresponding indices of the Green's functions in the LSZ reduction formula, and the final expression for the S-matrix elements will involve the appropriate spinor/vector structures for each type of particle in the in and out states:

$$\langle \Omega | T \{ \dots A^\mu(x) \dots \psi_\alpha(y) \dots \} | \Omega \rangle .$$

- For a Dirac fermion the ...

We have to make one final observation: we did not make any assumption on the behaviour of the field operators of the interacting theory, apart from the fact that they can be approximated by the field operators of the free theory in the asymptotic limit (as ladder operat), which is similar to the statement

$$\langle \Omega | \hat{\phi}(x) | \lambda_{\mathbf{p}} \rangle = \sqrt{Z} e^{-ip \cdot x},$$

[...]

Appendices

A | Notation and Conventions

Conventions on Units and Natural Units

We are going to use natural units throughout these notes, where the speed of light c and the reduced Planck constant $\hbar$ are set to 1:

$$c = 1, \quad \hbar = 1.$$

In natural units, all physical quantities are expressed in terms of a single unit of energy (or mass) and measured in Electronvolts (eV) and multiples thereof, since the dimensions of length and time are related to energy through the relations $E = mc^2$ and $E = \hbar\omega$. This simplifies many equations in quantum field theory and allows us to focus on the essential physics without being distracted by factors of c and $\hbar$. In these units, we have:

$$\begin{aligned} [E] &= [M] = [L]^{-1} = [T]^{-1}, \\ [\hbar] &= 1, \\ [c] &= 1. \end{aligned}$$

We report some useful conversion factors between natural units and SI units:

$$\begin{aligned} 1 \text{ GeV} &\approx 1.602 \times 10^{-10} \text{ J}, \\ 1 \text{ GeV} &\approx 1.783 \times 10^{-27} \text{ kg}, \\ 1 \text{ GeV}^{-1} &\approx 0.1975 \text{ fm}, \\ 1 \text{ GeV}^{-1} &\approx 6.582 \times 10^{-25} \text{ s}, \end{aligned}$$

and also

$$\begin{aligned} \hbar c &\approx 197.3 \text{ MeV} \cdot \text{fm}, \\ 1 \text{ fm} &\approx 10^{-15} \text{ m}, \end{aligned}$$

where the femtometer (fm) is a common unit of length in nuclear and particle physics, often called a Fermi: it is approximately the length of a proton or neutron radius:

$$r_p \sim 0.8 \text{ fm} = \frac{0.8}{197.3} \text{ MeV}^{-1}.$$

Conventions on Electromagnetism

In order to avoid confusion, we will use the **Heaviside-Lorentz** units for electromagnetism throughout these notes: in these units, the electric charge e is dimensionless and appears ex-

plicitly in the interaction terms of the Lagrangian, making it easier to identify the strength of electromagnetic interaction, while Maxwell equations take a more symmetric form.

Explicitly, in Heaviside-Lorentz units, we have the following relations:

- $\epsilon_0 = \mu_0 = 1$, where ϵ_0 is the *vacuum permittivity* and μ_0 is the *vacuum permeability*: this means that the electric and magnetic fields have the same units and that the speed of light c is equal to 1 (these units are consistent with the natural units we are using).
- The **electric charge** e is dimensionless and connected to the fine-structure constant α by

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{e^2}{4\pi} \approx \frac{1}{137}.$$

- The **electric potential** Φ is defined as

$$\Phi = \frac{Q}{4\pi R},$$

where Q is the charge and R is the distance from the charge: this means that the electric potential has the same units as the electric field E (which is consistent with the fact that in Heaviside-Lorentz units, $\epsilon_0 = 1$).

- The **Maxwell equations** take the following form:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \rho, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mathbf{J} + \frac{\partial \mathbf{E}}{\partial t},\end{aligned}$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ is the magnetic field, ρ is the charge density, and $\mathbf{J}$ is the current density: this symmetric form of Maxwell equations is one of the reasons why Heaviside-Lorentz units are often preferred in theoretical physics, especially in quantum field theory. Basically the Heaviside-Lorentz units are a rationalized version of the Gaussian units, where the factors of 4π are absorbed into the definitions of the fields and charges, leading to simpler equations and making it easier to identify the physical constants that govern the interactions.

Conventions on the Metric and Spacetime

We will use the **mostly minus** convention for the metric tensor $g_{\mu\nu}$ in Minkowski spacetime, which is defined as

$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

This means that the time component of the metric is positive, while the spatial components are negative; using Einstein summation convention, we have the following relations among the components of the metric and v^μ , w^μ four vectors:

$$\begin{aligned}v \cdot w &= g_{\mu\nu} v^\mu w^\nu = v^0 w^0 - \mathbf{v} \cdot \mathbf{w}, \\ v^2 &= v \cdot v = g_{\mu\nu} v^\mu v^\nu = (v^0)^2 - \mathbf{v}^2, \\ v_\mu &= g_{\mu\nu} v^\nu \implies v_0 = v^0, \quad \mathbf{v} = v^i = -v_i.\end{aligned}$$

We will use greek letters μ, ν, ρ, σ to denote spacetime indices that run from 0 to 3, while Latin letters i, j, k will be used to denote spatial indices that run from 1 to 3:

$$\begin{aligned}\mu, \nu, \rho, \sigma &\in \{0, 1, 2, 3\} = \{t, x, y, z\}, \\ i, j, k &\in \{1, 2, 3\} = \{x, y, z\}.\end{aligned}$$

Fourier Transforms

We will use the following conventions for Fourier transforms of a function $f(x)$ in four-dimensional spacetime:

$$\begin{aligned}f(x) &= \int \frac{d^4p}{(2\pi)^4} e^{-ip \cdot x} \tilde{f}(p), \\ \tilde{f}(p) &= \int d^4x e^{ip \cdot x} f(x).\end{aligned}$$

We highlight that the Fourier transform of a function $f(x)$ is denoted by $\tilde{f}(p)$, and that the integration measure includes a factor of $(2\pi)^{-4}$ in the inverse Fourier transform, which is a common convention in quantum field theory: in the space of coordinates, we have the measure d^4x , while in the momentum space, we have the measure $\frac{d^4p}{(2\pi)^4}$. This convention ensures that the Fourier transform and its inverse are consistent and that the normalization of the fields and operators in quantum field theory is correct.

Feynman Diagrams

In these notes, we will use Feynman diagrams as a graphical representation of the interactions between particles and fields in quantum field theory. We will use the following conventions for Feynman diagrams:

TODO: Check and change in the end

- Time will be represented on the horizontal axis, going from left to right, which is a common convention in quantum field theory; this means that the left side of the diagram will represent the initial state of the particles, while the right side will represent the final state after the interaction has occurred.
- Solid lines will represent fermions (spin-1/2 particles), while dashed lines will represent bosons (spin-0 or spin-1 particles).
- Arrows on the lines will indicate the direction of particle flow, with arrows pointing towards the interaction vertex for incoming particles and away from the vertex for outgoing particles.
- Wavy lines will represent gauge bosons (spin-1 particles that mediate interactions), such as photons, W and Z bosons, and gluons.
- The vertices in the diagrams will represent interaction points where particles interact according to the rules of the theory, with each vertex corresponding to a specific term in the Lagrangian of the quantum field theory.

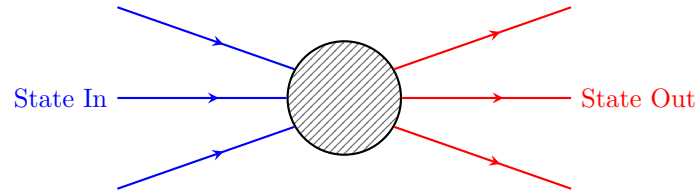


Figure 5: Generic Feynman diagram representing an interaction between particles, where the blue lines represent incoming particles (initial state) and the red lines represent outgoing particles (final state). The black box represents the interaction vertex, which corresponds to a specific term in the Lagrangian of the quantum field theory.

B | Computations and further developments

Differential Cross Section

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Decay Rate

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